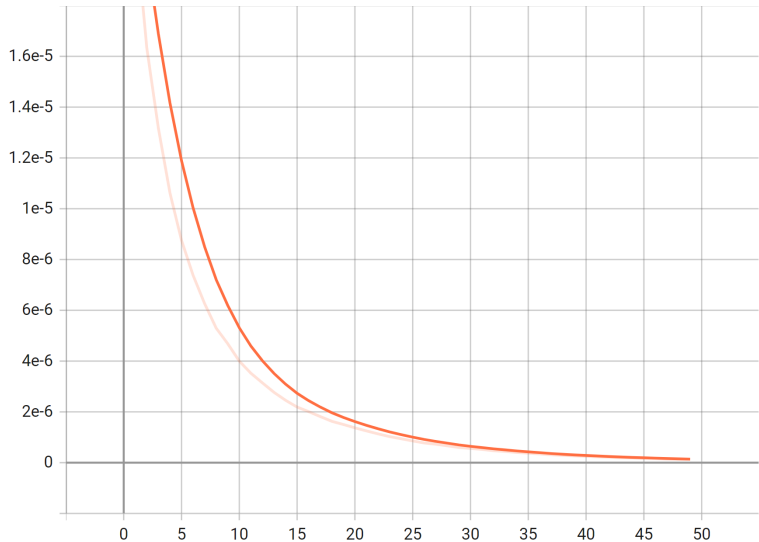
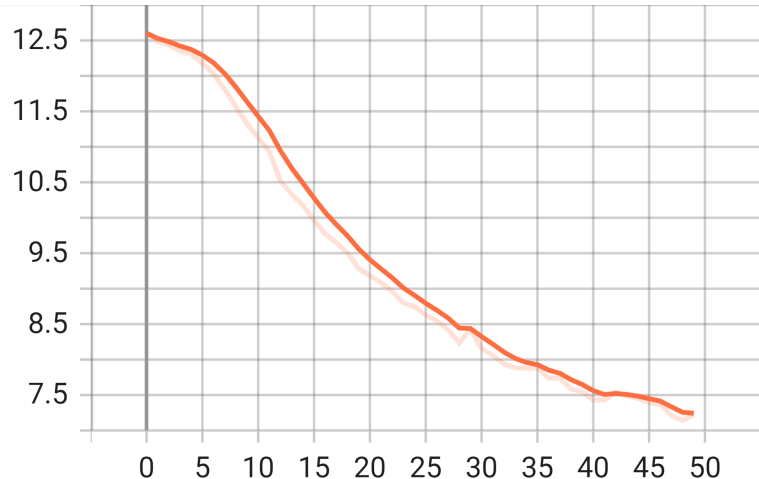
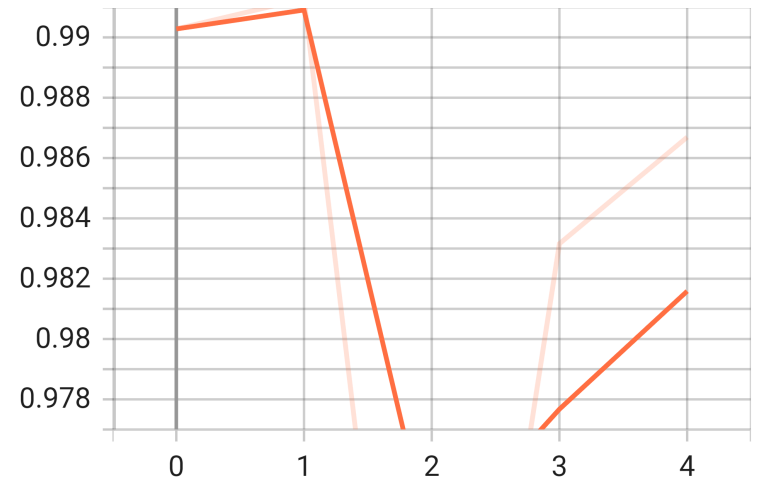


Deep Learning Final Project Report

Loss Curves:

Function	x / y	Chart																								
Discriminator Loss	Discriminator Loss / Epoch	 This chart displays the Discriminator Loss over 50 epochs. The y-axis represents the loss value, ranging from 0 to 1.6e-5 with increments of 2e-6. The x-axis represents the epoch number, ranging from 0 to 50 with increments of 5. A solid red line shows the loss starting at approximately 1.6e-5 at epoch 0 and decreasing rapidly, reaching near zero by epoch 20. A light red shaded area indicates the variance or confidence interval around the mean loss. <table><tr><th>Epoch</th><th>Discriminator Loss</th></tr><tr><td>0</td><td>1.6e-5</td></tr><tr><td>5</td><td>1.0e-5</td></tr><tr><td>10</td><td>6e-6</td></tr><tr><td>15</td><td>3e-6</td></tr><tr><td>20</td><td>1.5e-6</td></tr><tr><td>25</td><td>8e-7</td></tr><tr><td>30</td><td>4e-7</td></tr><tr><td>35</td><td>2e-7</td></tr><tr><td>40</td><td>1e-7</td></tr><tr><td>45</td><td>5e-8</td></tr><tr><td>50</td><td>2e-8</td></tr></table>	Epoch	Discriminator Loss	0	1.6e-5	5	1.0e-5	10	6e-6	15	3e-6	20	1.5e-6	25	8e-7	30	4e-7	35	2e-7	40	1e-7	45	5e-8	50	2e-8
Epoch	Discriminator Loss																									
0	1.6e-5																									
5	1.0e-5																									
10	6e-6																									
15	3e-6																									
20	1.5e-6																									
25	8e-7																									
30	4e-7																									
35	2e-7																									
40	1e-7																									
45	5e-8																									
50	2e-8																									
Generator Loss (MSE)	Generator Loss / Epoch	 This chart displays the Generator Loss (MSE) over 50 epochs. The y-axis represents the loss value, ranging from 7.5 to 12.5 with increments of 1.0. The x-axis represents the epoch number, ranging from 0 to 50 with increments of 5. A solid red line shows the loss starting at 12.5 at epoch 0 and decreasing steadily to approximately 7.2 by epoch 50. A light red shaded area indicates the variance or confidence interval around the mean loss. <table><tr><th>Epoch</th><th>Generator Loss (MSE)</th></tr><tr><td>0</td><td>12.5</td></tr><tr><td>5</td><td>12.2</td></tr><tr><td>10</td><td>11.5</td></tr><tr><td>15</td><td>10.5</td></tr><tr><td>20</td><td>9.5</td></tr><tr><td>25</td><td>8.8</td></tr><tr><td>30</td><td>8.2</td></tr><tr><td>35</td><td>7.8</td></tr><tr><td>40</td><td>7.5</td></tr><tr><td>45</td><td>7.3</td></tr><tr><td>50</td><td>7.2</td></tr></table>	Epoch	Generator Loss (MSE)	0	12.5	5	12.2	10	11.5	15	10.5	20	9.5	25	8.8	30	8.2	35	7.8	40	7.5	45	7.3	50	7.2
Epoch	Generator Loss (MSE)																									
0	12.5																									
5	12.2																									
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15	10.5																									
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30	8.2																									
35	7.8																									
40	7.5																									
45	7.3																									
50	7.2																									
Generator Loss / adversarial loss	Generator Loss / Adversarial Loss	 This chart displays the ratio of Generator Loss to Adversarial Loss over 4 epochs. The y-axis represents the ratio value, ranging from 0.978 to 0.99 with increments of 0.002. The x-axis represents the epoch number, ranging from 0 to 4 with increments of 1. A solid red line shows the ratio starting at approximately 0.99 at epoch 0, peaking at 0.99 at epoch 1, then dropping sharply to approximately 0.978 at epoch 2. A light red shaded area indicates the variance or confidence interval around the mean ratio. <table><tr><th>Epoch</th><th>Generator Loss / Adversarial Loss</th></tr><tr><td>0</td><td>0.99</td></tr><tr><td>1</td><td>0.99</td></tr><tr><td>2</td><td>0.978</td></tr><tr><td>3</td><td>0.983</td></tr><tr><td>4</td><td>0.987</td></tr></table>	Epoch	Generator Loss / Adversarial Loss	0	0.99	1	0.99	2	0.978	3	0.983	4	0.987												
Epoch	Generator Loss / Adversarial Loss																									
0	0.99																									
1	0.99																									
2	0.978																									
3	0.983																									
4	0.987																									

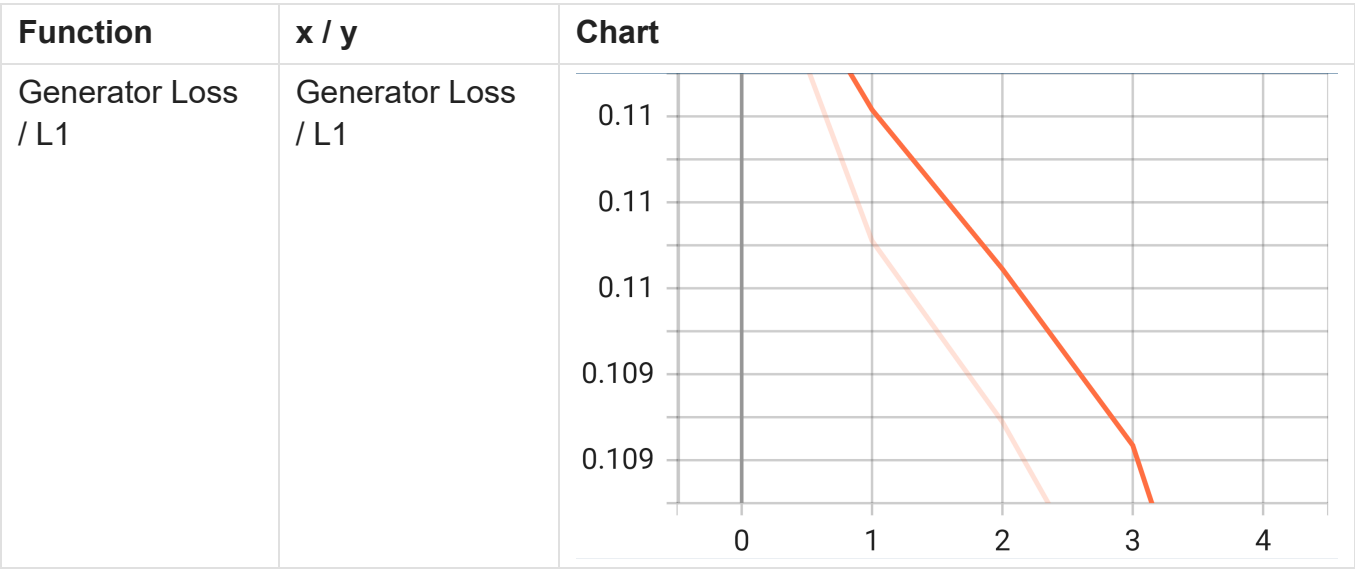
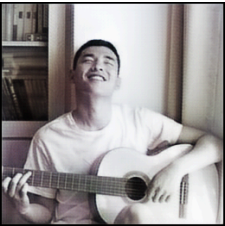
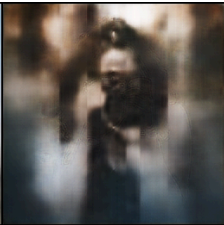
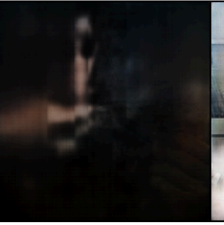
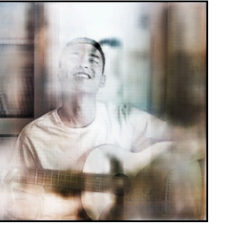


Image Outputs

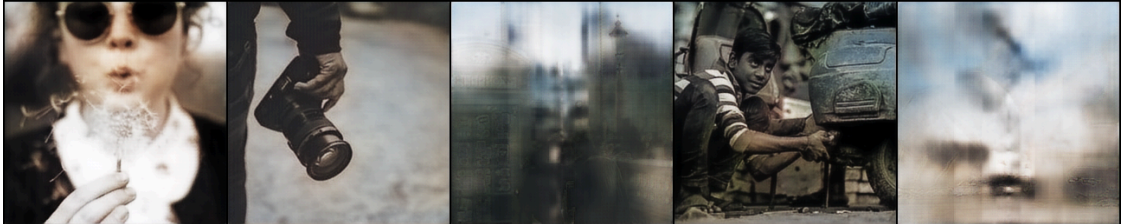
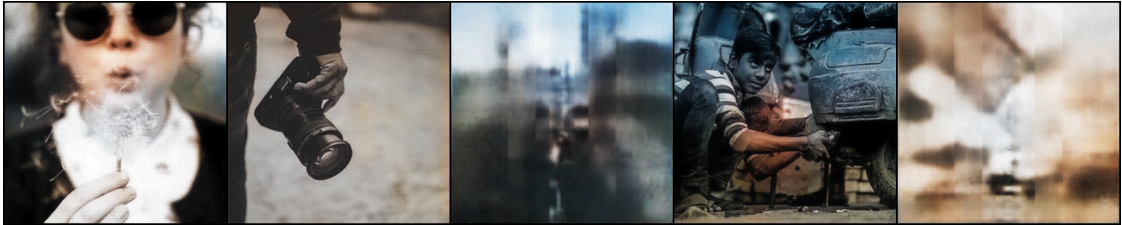
Testing Images:

Generation	Output
Input	Training Images 
Epoch 0	
Epoch 5	

Generation	Output
Epoch 10	    
Epoch 22	    
Epoch 45	    
Epoch 48	Training Images     

Training Images:

Generation	Output
Epoch 0	    
Epoch 5	    

Generation	Output
Epoch 10	
Epoch 22	
Epoch 45	
Epoch 48	
Actual	